

Experimental Outcome

Growth (+)

No Growth (—)

GapMind Prediction

Pathway Present (+)

Pathway Absent (—)

CONCORDANT

True Positive

Genes present and functional
Pathway correctly annotated
Experiment accurately measured

High-quality training samples

DISCORDANT

False Negative

Possible Causes:

- Annotation: homolog not detected
- Biology: alternative pathway
- Biology: promiscuous enzymes
- Measurement: false positive
- Media: growth on base media

Growth via unknown mechanism

DISCORDANT

False Positive

Possible Causes:

- Regulation: pathway repressed
- Measurement: insufficient time
- Media: missing cofactors
- Annotation: false positive call
- Biology: incomplete pathway

Genes present but not functional

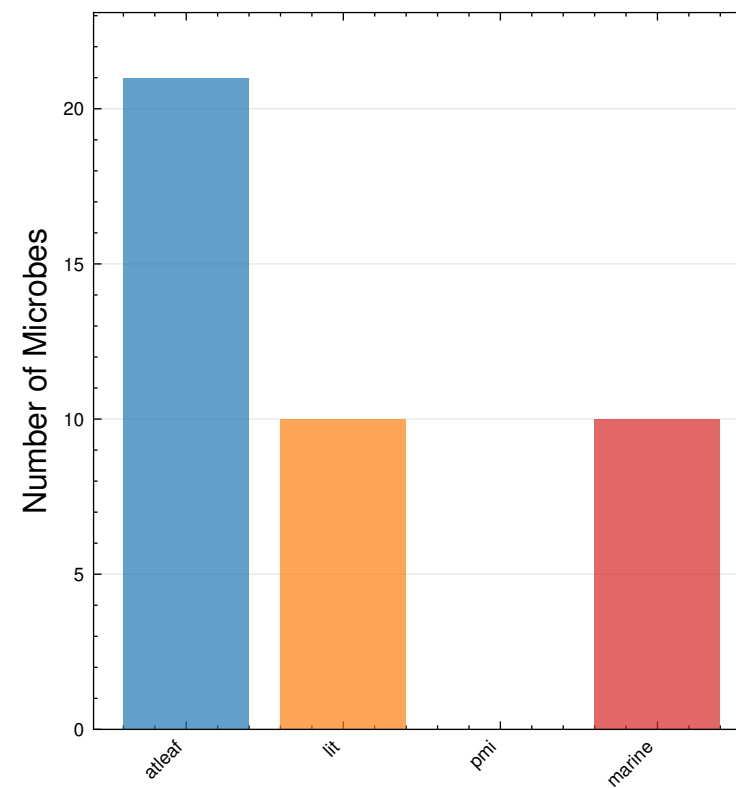
CONCORDANT

True Negative

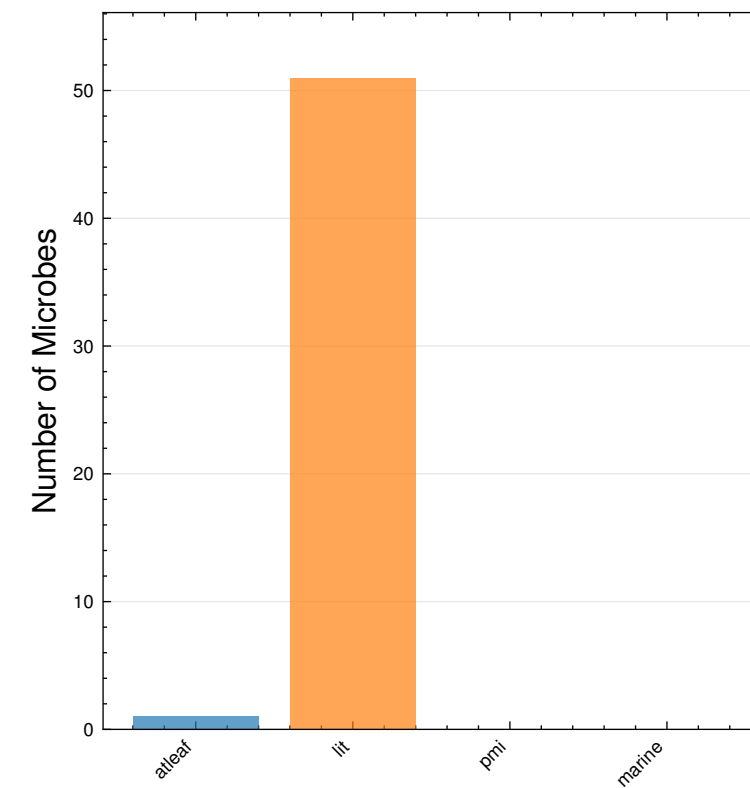
Genes absent
No alternative pathway
Experiment accurately measured

High-quality training samples

No Experimental Growth but GapMind Predicts Growth



Growth on All C Sources but GapMind Incomplete



Dataset

Dataset

Top 20 Most Frequently Misclassified Microbes

No Growth All Growth Neither

Microbe ID

